Abdelrahman Mostafa

AI Engineer | Generative AI, LLMs & RAG

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PROFESSIONAL SUMMARY

Recent Cairo University graduate focused on LLM applications, retrieval-augmented generation (RAG), and evaluation. Led teams on a healthcare Q&A assistant that declines when its sources are weak and on a LangGraph travel planner.

TECHNICAL SKILLS

- **Generative AI & Agents:** RAG, agentic workflows, LangGraph, LangChain, prompt engineering, LLM APIs (Gemini, Groq, Cerebras)
- **Retrieval & Evaluation:** embeddings (BGE-M3, sentence-transformers), vector search (FAISS), cross-encoder reranking, LLM-as-judge evaluation, citation validation, evidence-aware abstention
- **Machine Learning:** scikit-learn, XGBoost, Pandas, NumPy, feature engineering, cross-validation, SMOTE
- **Backend & DevOps:** FastAPI, REST APIs, PostgreSQL, Redis, pytest, Docker, GitHub Actions (CI), MLflow, Git
- **Programming:** Python, SQL, C++

EXPERIENCE

ML Engineer Trainee | DEPI (Microsoft ML Track) *Nov 2025 – Jul 2026*

- Led a 5-person Agile team through nine months of sprints (planning, task allocation, code reviews) to deliver the Healthcare RAG Assistant capstone.
- Owned the FastAPI backend and CI (GitHub Actions) running **216 automated tests** on every push and pull request.

Machine Learning Intern | Elevvo Pathways *Sep 2025 – Oct 2025*

- Built **four** end-to-end ML projects across regression, clustering, and classification: exam-score prediction, customer segmentation, loan approval, and forest-cover classification.
- Applied data cleaning, feature engineering, cross-validation, and SMOTE where appropriate, using suitable metrics.

PROJECTS

TourMate AI | Agentic Travel Planning Platform (Cairo University Graduation Project) | Demo *2025 – 2026*

- Led a 6-member team building a travel-planning platform (FastAPI, PostgreSQL, Redis, Flutter) with Amadeus flight search and Stripe payments.
- Co-built a 6-node LangGraph pipeline that turns chat requests into multi-day itineraries: profile loading, retrieval over **6,700+ places** in four Egyptian cities, candidate ranking, LLM planning, OSRM-based route optimization, and validation with retries.
- Integrated Gemini 2.5 Flash and Groq Llama 3.3 70B with provider fallback, key rotation, and rate-limit retries.

Healthcare RAG Assistant | Evidence-Grounded Health Q&A (DEPI Capstone) | Demo *2025 – 2026*

- Built a health Q&A assistant over MedQuAD, MedlinePlus, and USPSTF using BGE-M3, FAISS retrieval, cross-encoder reranking, and Gemini; it answers only when retrieved evidence is sufficient and abstains otherwise.
- Added rule-based and Gemini-based safety screening, with abstention for questions requiring up-to-date information.
- Evaluated on a **255-question** labeled dataset; on its **87-question** test subset, matched the expected behavior on **79 questions (90.8%)**. Results are exploratory, and labels are not clinician-reviewed.

EDUCATION

B.Sc. in Computers and Artificial Intelligence | Cairo University | GPA: 3.53 / 4.00 | Graduated Jul 2026

CERTIFICATIONS

- Machine Learning Specialization – Stanford University / Andrew Ng, Coursera
- Building LLM Applications with Prompt Engineering – NVIDIA Deep Learning Institute
- Associate Data Scientist in Python – DataCamp

ACHIEVEMENTS

- **Competitive Programming:** Solved 550+ problems across Codeforces, LeetCode, and HackerRank.
- **ICPC / ECPC Regionals:** 84th at ECPC 2023 Regionals; Honorable Mention at ECPC 2025 Regionals (ICPC ID).

LANGUAGES

Arabic (Native) | English (Professional)